

CSE 318 Assignment-03: Adversarial Search Report

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Experimentation & Analysis

Experimental Setup

The experiment evaluated a minimax agent with alpha-beta pruning in the Chain Reaction game on a 9x6 board, using heuristics 1-7 defined in `heuristics.hpp`. Three experiment types were conducted:

- **AI vs. Human-like AI:** AI with heuristics 1-7 at depths 1-3 played against a human-like AI (heuristic 2, depth 1). Each configuration ran 10 games, with AI1 as 'R' (red) and human as 'B' (blue).
- **Random vs. AI:** A random agent (no heuristic or depth, denoted as -1) faced AI with heuristics 1-7 at depth 3, over 10 games per heuristic. AI was 'R', and the random agent was 'B'.
- **AI vs AI:** AI with heuristics 1-7 at depths 1-3 played against another AI with heuristics 1-7 and depths 1-3. Each configuration ran 10 games, with AI1 as 'R' (red) and AI2 as 'B' (blue)

Results

Table 1 summarizes key results from `experiment_results.csv`, highlighting strong performances.

Table 1: Results for AI vs. Human-like AI and Random vs. AI

Experiment Type	Heuristic AI1	Depth AI1	Heuristic AI2	Depth AI2	AI1 Win Rate (%)	Avg Duration (s)
AI vs. Human	6	1	2	1	100.0	0.1
AI vs. Human	6	3	2	1	100.0	7.9
AI vs. Human	3	1	2	1	100.0	0.0
AI vs. Human	3	3	2	1	100.0	4.3
AI vs. Human	2	1	2	1	0.0	0.1
AI vs. Human	2	3	2	1	100.0	15.9
AI vs. AI	7	2	1	2	100.0	13.6
AI vs. AI	3	3	6	3	0.0	579.8
AI vs. AI	3	2	6	2	100.0	137.5
Random vs. AI	6	3	-1	-1	100.0	41.1
Random vs. AI	3	3	-1	-1	100.0	13.3
Random vs. AI	2	3	-1	-1	70.0	11.8

Discussion

In *AI vs. Human-like AI* heuristic 6 achieved a 100% win rate at depth 1 (0.1s avg duration), showcasing its efficiency, and maintained this at depth 3 (7.9s). Heuristic 3 also excelled, with 100% win rates at depths 1 (0.0s) and 3 (4.3s), indicating strong performance with minimal computation at lower depths. Heuristic 2 struggled at depth 1 (0% win rate) but reached 100% at depth 3, though with a longer duration (15.9s), suggesting depth dependency.

In "Random vs. AI," heuristic 6 again led with a 100% win rate at depth 3 (41.1s), followed by heuristic 3 (100%, 13.3s). Heuristic 2 lagged with a 70% win rate (11.8s), underscoring its weaker performance against even a random agent. The trade-off between depth and duration is evident, with heuristic 6's complexity increasing computation time.

Analysis of AI vs. Other AI Matchups

Experimental Setup

For Task 5, the game engine was adapted for AI vs. AI matches, pitting each heuristic (1-7) at depths 1-3 against all other heuristics at the same depth. Each configuration ran 6 games per opponent, with results aggregated in `experiment_results2.csv`. AI1 was 'R', and opponents were 'B'. Metrics mirrored Task 4.

Results

Table 2 summarizes `experiment_results2.csv`, focusing on heuristics 6, 3, and 2.

Table 2: Aggregated AI vs. AI Results

Heuristic AI1	Depth AI1	AI1 Win Rate (%)	Avg Duration (s)
6	1	83.3	0.1
6	2	100.0	4.8
6	3	100.0	82.4
3	1	50.0	0.0
3	2	100.0	2.5
3	3	66.7	46.6
2	1	16.7	0.0
2	2	0.0	1.3
2	3	0.0	40.0

Discussion

Heuristic 6 dominated in "AI vs. AI," with an 83.3% win rate at depth 1 (0.1s) and 100% at depths 2 (4.8s) and 3 (82.4s), reflecting its robust evaluation of corners, edges, and explosion chains. Heuristic 3 ranked second, with 50% at depth 1 (0.0s), 100% at depth 2 (2.5s), and 66.7% at depth 3 (46.6s), leveraging its focus on immediate threats. Heuristic 2 performed worst, with 16.7% at depth 1 (0.0s) and 0% at depths 2 and 3, despite increasing durations (40.0s at depth 3), due to its simplistic neighbor-based approach.

The trade-off between performance and computation is stark: heuristic 6's high win rates come with significant time costs at depth 3, while heuristic 3 balances efficiency and effectiveness, making it viable for faster play.